

Multiple Choice (10 marks)

1. Using conditioning, how can one determine the sensitivity of a dog's hearing to sounds of different frequencies?
 - (a) Through ultrasound examination
 - (b) Through blood sample analysis
 - (c) Through direct observation
 - (d) Through measuring brain wave activity in response to sounds
 - (e) Through operant conditioning
2. The volume of plasma and other extracellular fluids are regulated by automatic feedback controls. What are these controls?
 - (a) The controls are nerve impulses in the body
 - (b) The controls are the contraction and relaxation cycles of skeletal muscles
 - (c) The controls are hormonal responses that regulate digestion and absorption of food
 - (d) The controls are mechanisms that regulate the heart rate
 - (e) The controls are processes that control the body temperature
3. What are the typical periods in the unfolding of a behavioral act?
 - (a) Appetitive behavior, exploratory behavior, quiescence
 - (b) Termination, appetitive behavior, exploratory behavior
 - (c) Appetitive behavior, consummatory act, quiescence
 - (d) Consummatory act, exploratory behavior, termination
 - (e) Appetitive behavior, termination, consummatory act
4. Based on the characteristic population curves that result from plotting population growth of a species, the most effective means of controlling the mosquito population is to
 - (a) opt for zero population control once the K value of the curve has been reached
 - (b) maintain the population at the highest point of its logistic curve
 - (c) reduce the carrying capacity of the environment to lower the K value
 - (d) decrease the mortality rate
 - (e) increase the birth rate of the species
5. Two related species evolve in a way that makes them less similar.

- (a) Coevolution
- (b) Punctuated equilibrium
- (c) Directional selection
- (d) Divergent evolution
- (e) Convergent evolution

True / False (10 marks)

Answer True or False

- 6. True or False: Is there still a need for living-related liver transplantation in children?
- 7. True or False: Should displaced midshaft clavicular fractures be treated surgically?
- 8. True or False: The colour of pain: can patients use colour to describe osteoarthritis pain?
- 9. True or False: Does elective re-siting of intravenous cannulae decrease peripheral thrombophlebitis?
- 10. True or False: Do Wound Cultures Give Information About the Microbiology of Blood Cultures in Severe Burn Patients?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the roles of cortisol, aldosterone, and adrenal androgens in the physiological changes that occur during puberty in both males and females.
- 12. Discuss the potential consequences on cellular function and gene expression if messenger RNA molecules were not degraded after polypeptide synthesis.

End of Paper